Esercizio 1. (Martina Apicella & Beatrice Calderaro) Let M be ω-saturated. Let $A \subseteq M$ be finite. Prove M realizes every consistent $p(x) \subseteq L(\mathrm{acl}A)$, for $|x| < \omega$.

Esercizio 2. (Francesco Dam & Daniele Lanciotti) Prove that if T is ω-categorical then for every $n < \omega$ there is an $m < \omega$ such that $|\mathrm{acl}A| \leq m$ for every A such that $|A| \leq n$.

Esercizio 3. (Filippo Giugie) Let $p(x) \subseteq L(A)$ and let $\varphi(x;y) \in L(A)$ be a formula that defines, when restricted to $p(\mathfrak{U}^x)$, an equivalence relation with finitely many classes. Prove that there is a finite equivalence relation definable over A that coincides with $\varphi(x;y)$ on $p(\mathfrak{U}^x)$.

Esercizio 4. (Mattia Lo Mele & Andrea Morello) Assume that T is strongly minimal. Let $\varphi(x;z) \in L(A)$ where $|x| = 1$. For arbitrary $b \in \mathfrak{U}^z$, prove that if the orbit of $\varphi(\mathfrak{U}^x;b)$ over A is finite, then $\varphi(\mathfrak{U}^x;b)$ is definable over $\mathrm{acl}A$.